

The n-adic Collatz Universe

— A General Framework of Integer Dynamics through Gravity and Modular Jumps —

1. Overview

The classical Collatz conjecture (the “ $3x+1$ problem”) is a special case of a broader class of integer dynamical systems. This work introduces the **n-adic Collatz Universe**, a general framework where each integer base n ($n \geq 2$) defines a unique numerical cosmos.

In this universe, two alternating phases govern the evolution of integers:

- A **Gravity Phase**, where division by proper divisors of n pulls values downward.
- A **Jump Phase**, where coprime values accelerate *via* modular jumps.

This structure reveals that convergence, cycles, and divergence are not fixed properties, but tunable behaviors shaped by the parameters of each universe.

2. Core Mechanics

• Gravity Phase (ARD: Adaptive Recursive Division)

If x is divisible by any proper divisor d of n (where $d \geq 2$), divide x by the largest such d . This simulates gravitational collapse toward the core (value 1).

• Jump Phase (Modular Acceleration)

If x is coprime to n , apply the jump function:

$$f(x) = ((n + 1) \times x + c_r) / n \quad \text{where } r = x \bmod n, \text{ and } c_r \text{ is chosen so that } r + c_r \equiv 0 \bmod n$$

This ensures that $f(x)$ is always an integer. The constants c_r define the “jump energy” for each residue class.

3. Case Studies: 4-adic Universe (Collatz-4)

Gravity Wells and Asymmetric Jumps

In the 4-adic Collatz Universe, the dynamics are shaped by the divisors 4 and 2, creating strong gravitational pulls, while modular jumps inject bursts of acceleration based on residue classes. With modular constants set as $c_1 = -1$ and $c_3 = +1$, the system exhibits a fascinating interplay of descent and surge.

Example: 20,737

Starting from 20,737, the orbit undergoes a dramatic sequence of transformations. Despite frequent collapses via division by 4 and 2, the number experiences potent modular jumps that propel it to a **maximum of 150,876**. After **41 steps**, the orbit ultimately converges to 1. This trajectory exemplifies a high-altitude oscillation before gravitational stabilization—a hallmark of 4-adic turbulence.

4. Insights

- The $3x+1$ problem is just one instance in the broader n -adic family.
- The behavior of each universe is shaped by the balance between **gravitational division** and **modular acceleration**.
- By tuning n and the constants c_r , one can design stable, chaotic, or divergent systems.

5. Tools and Methodology

- Python-based simulation engine
- Hash-set cycle detection
- Divergence ceiling at 10^{18}
- Batch scanning for $1 \leq x \leq 100$
- Interactive lab mode for custom universe creation

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